Plasmodium Species Identification by ELISA for Sporozoites Removed from Dried Dissection Slides

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ABSTRACT This study tested the feasibility of identifying salavary gland sporozoites to species by *Plasmodium falciparum* ELISA by drying them on slides or in vials. The glands were dissected from *Anopheles gambiae* Giles s.l. and *An. funestus* Giles collected in western Kenya. In 119 gland infections containing a geometric mean of 1,222 sporozoites, a mean of 72.5% of sporozoites were removed in 60 μl saline from slides at the time of dissection. Each of the 119 samples was divided into three 18 μl aliquots. Subsamples were stored at −70°C, dried in vials, or dried on a microslide. When tested by *Plasmodium falciparum* ELISA, positive reactions were observed in 86.6% of frozen samples, 70.6% of samples held dry in vials, and 50.4% of samples held dry on microslides for 1 mo. Of 90 gland infections where coverslips were removed and slides were left to dry for 1 mo before adding blocking buffer, 81.1% were positive for *P. falciparum*. This was not significantly different from either frozen gland samples (where 85.5% of 392 infections were identified or frozen gland plus corresponding thorax samples where 86.2% of 160 samples were identified). In malaria field studies, where it is not always practical to freeze samples, sporozoites from dissected mosquitoes can be preserved adequately for ELISA identification by simply removing coverslips and drying dissection slides.

KEY WORDS Insecta, *Anopheles*, sporozoites, ELISA

IMMUNOASSAYS FOR MALARIA SPOROZOITES have been used to estimate field infection rates in *Anopheles* populations and to identify the species of *Plasmodium* sporozoites removed from dissection slides. In conjunction with dissections, sporozoites and gland material may be removed from slides and frozen before testing by *Plasmodium*-specific enzyme-linked immunosorbent assays (ELISA) (Beier et al. 1987, 1989, 1991; Delorme et al. 1989) or by immunoradiometric methods (Lombardi et al. 1987). In western Kenya, this method was used to identify 92.5% of *P. falciparum*, *P. malariae*, *P. ovale*, and mixed salivary gland infections in 1,113 Afrotropical *Anopheles* (M.S. Beier et al. 1988).

Our current study examines the possibility that sporozoites detected by dissection may be air-dried either in vials or on dissection slides and stored for later testing by ELISA. Establishment of the ELISA detection limits for known numbers of dried sporozoites was necessary, in light of evidence that circumsporozoite protein remains intact in dried whole mosquitoes (Zavala et al. 1982), and because of the variability in sporozoite loads of field-collected anophelines (Beier et al. 1991). The sensitivity of the *P. falciparum* ELISA was compared with counted sporozoites from salivary gland infections that were divided into three aliquots and held either frozen, dry in vials, or dry on microslides. The feasibility of testing dried sporozoites in malaria field studies was demonstrated by comparing rates of sporozoite identification by ELISA for infected mosquitoes where sporozoites and gland material were either left dry on dissection slides, frozen in vials, or frozen in vials along with corresponding thoraces.

Materials and Methods

Anopheles gambiae s.l. Giles and *An. funestus* Giles were collected by all-night human-biting catches and by daytime aspiration collections inside houses from the villages of Kisian and Saradidi, western Kenya (Beier et al. 1990b). Sporozoites in the salivary glands were detected during routine dissections on the day of collection and were graded as to the number of sporozoites observed: +1 (1–10), +2 (11–100), +3 (101–1,000) and +4 (>1,000).

During June 1988, sporozoites from 119 gland infections were removed from standard dissection slides in 60 μl of physiological saline and placed in 2.0-ml polystyrene vials. The sporozoite content of each 60-μl sample was determined as the number of sporozoites counted in a 5-μl aliquot (placed on a microslide and stained with Giemsa) times 12 to

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account for the total volume. In conjunction, sporozoites remaining on each dissection slide were counted after staining with Giemsa. Thus, sporozoite loads for each infected mosquito were calculated as the number of sporozoites removed from slides in 60-μl samples, plus the number of sporozoites counted on original dissection slides (Beier et al. 1989).

A *P. falciparum* sporozoite ELISA (Wirtz et al. 1987b) was used to test three 18-μl aliquots which were held either: frozen at −20°C in a 2.0-ml vial, dry in a 2.0-ml vial, or dry on a microslide. Desiccant was not used to dry samples. Dried samples were exposed for 3 d at room temperature (20–27°C) before capping the vials and before slides were stored in slide boxes. After 1 mo, all samples were mixed with 82 μl blocking buffer containing NP40 (Wirtz et al. 1987b) and then frozen at −20°C before ELISA testing. For the sporozoites held dry on the microslide, the blocking buffer was added to the slide, incubated for 15 min, then mixed and removed with a pipette. Fifty microliters of each sample were tested by ELISA. Samples were considered positive if absorbance values exceeded twice the mean of four negative controls per microtiter plate (Beier et al. 1988); $R32tet_{32}$ was used as a positive control (Wirtz et al. 1987a).

The proportion of *P. falciparum* sporozoite infections was determined by ELISA for a series of individual mosquitoes where: (1) sporozoites were held dry on dissection slides (with coverslips removed) for 1 mo, (2) sporozoites were removed from slides in 60 μl saline and held frozen, or (3) sporozoites removed from slides in 60 μl saline were mixed with corresponding thoraces and held frozen (these were ground before freezing). The total volume of all samples was adjusted to 150 μl by the addition of blocking buffer. For the sporozoites held on dissection slides, the blocking buffer was incubated for 15 min before being removed and placed in a vial. Fifty-microliter samples were tested by the *P. falciparum* ELISA as described above.

Results

Salivary gland sporozoites were detected in 109 *An. gambiae* s.l. and in 10 *An. funestus*. Infections ranged from 1 to 79,732 sporozoites, with a geometric mean of 1,222 sporozoites per female; there were no significant differences in sporozoite loads between *An. gambiae* s.l. and *An. funestus* (F = 2.24; df = 1,117; P = 0.137). Less than 100 sporozoites were detected in 8.4% (10/119) of the infections and more than 10,000 sporozoites were detected in 11.8% (14/119).

An average of 72.5% of the sporozoites was removed from dissection slides. There was a positive correlation between the proportion of sporozoites removed from slides and sporozoite loads (r = 0.41, df = 117, P < 0.0001). Over 100 sporozoites were removed from 84.9% (101/119) of the samples.

The ELISA identification rate for *P. falciparum* for sporozoites removed from slides in 60 μl of saline and divided into three 18-μl aliquots was 86.6% for frozen samples, 70.6% for sporozoites held dry in a vial, and 50.4% for sporozoites held dry on microscope slides (χ^2 = 36.46, df = 2, P < 0.01) (Table 1). In each of the test categories, both the rate of identification and ELISA absorbance increased as a function of the number of sporozoites. For the 16 frozen samples that were ELISA negative, dry and slide samples were correspondingly negative.

Positive ELISA reactions were observed for 81.1% of 90 sporozoite samples held dry on dissection slides before testing (Table 2). Mean absorbance values ranged from 0.76 for +1 to 1.36 for +4 glands, averaging 1.15. The rate of ELISA identifications for sporozoites from dry slides (81.1%) was not significantly different from the 85.5% (n = 392) rate of identification for frozen gland samples (χ^2 = 0.94, df = 1, P > 0.05) or the 86.2% (n = 160) rate of identification for frozen glands plus thorax (χ^2 = 1.19, df = 1, P > 0.05). Overall, there were no significant differences among the three holding conditions (χ^2 = 1.72, df = 2, P > 0.05).

Discussion

The methods used for determining sporozoite loads proved useful for estimating the number of sporozoites removed from dissection slides. Because >90% of the gland infections contained >100 sporozoites, the removal of an average of 72.5% of the sporozoites was satisfactory for species identification by ELISA; the *P. falciparum* sporozoite ELISA can detect as few as 25 sporozoites per test sample (Wirtz et al. 1987b). The geometric mean of 1,222 sporozoites per gland infection was similar to a more extensive study in western Kenya (Beier et al. 1991) and indicated that tests were made on sporozoite infections that were representative for this area.

Dividing sporozoites from the salivary glands of single mosquitoes into three aliquots indicated that ELISA sensitivity was greatest when samples were held frozen compared with holding them either dry in a vial or dry on microslides. For all three test conditions, both the proportion identified and ELISA absorbance increased as a function of the number of sporozoites per sample. When >100 sporozoites were held dry in a vial for 1 mo, >85% of the infections were identified. The slightly lower sensitivity for sporozoites held dry on microslides was probably caused by a loss of sporozoite material associated with the removal of blocking buffer from the slide.

The feasibility of simply removing coverslips from dissection slides and storing slides dry before ELISA testing was demonstrated by identifying 81.1% of 90 gland infections. This was similar to the 392 frozen samples, where 85.5% were identified by ELISA. In 160 additional samples, it was

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Table 1. Identification of sporozoites removed from dissection slides in 60 μl saline and divided into 18-μl aliquots which were either held frozen, dry in vials, or dry on slides, then tested using a *P. falciparum* ELISA

No. sporozoites/50-μl sample[a]	Frozen				Dry in vial			Dry on slide		
	n	% Positive	Absorbance		% Positive	Absorbance		% Positive	Absorbance	
			$\bar{x}$	SD		$\bar{x}$	SD		$\bar{x}$	SD
0–10	15	73.3	0.83	0.62	40.0	0.23	0.18	33.3	0.24	0.21
11–100	43	81.4	0.81	0.54	55.8	0.39	0.39	32.6	0.23	0.27
101–1,000	36	91.7	1.31	0.57	86.1	0.94	0.61	55.6	0.51	0.49
>1,000	25	96.0	1.57	0.50	92.0	1.43	0.67	84.0	1.03	0.58
Total	119	86.6	1.28	0.52	70.6	1.02	0.63	50.4	0.84	0.51

[a] The number of sporozoites in each test sample was based on sporozoite counts made on 5-μl aliquots of the original 60 μl removed from dissection slides.

not possible to increase the proportion of infections identified by the *P. falciparum* ELISA by testing both frozen gland and thorax. This was surprising because the thorax of infective mosquitoes, after removal of salivary glands, contains significant amounts of circumsporozoite protein (Beier et al. 1990a).

When testing sporozoites removed from dissection slides by ELISA, it usually is not possible to identify 100% of infections. For example, Delorme et al. (1989) identified only 42.5% of gland infections in Malaysia. In western Kenya, the consistent identification of 80–90% of the sporozoite infections as *P. falciparum* (Beier et al. 1987, 1991) raises the question of the identity of 10–20% of the infections. Only about 5% of the sporozoite infections have been identified as *P. malariae* or *P. ovale* (M.S. Beier et al. 1988). The failure to identify some gland sporozoite infections, despite high sporozoite loads, may be related to conformational changes in the repeat region caused by interactions with glandular secretions (Golenda et al. 1990), to epitope variation as demonstrated for *P. vivax* sporozoites (Rosenberg et al. 1989), or to technical problems of using several ELISAs with varying degrees of sensitivity.

The implications of these results for malaria field studies is that sporozoites from dissection slides do not have to be removed immediately from slides and held frozen for testing by immunoassays. Circumsporozoite protein remains intact on dried dissection slides for at least 1 mo as demonstrated in this study, and probably much longer as indicated by preliminary tests on dissection slides held 1 yr when most sporozoites were removed at the time of dissection (J.C.B., unpublished data). These findings may be especially valuable for situations in which dissections are performed in remote areas where it is not possible to freeze samples.

Table 2. *Plasmodium falciparum* ELISA identification of dissected salivary gland sporozoites held either dry on dissection slides (dry slide), removed from slides and stored frozen (frozen gland), or removed from slides, mixed with the thorax, and stored frozen (frozen gland and thorax)

Salivary gland grade	Dry slide		Frozen gland		Frozen gland and thorax	
	n	% Positive	*n*	% Positive	*n*	% Positive
+1	8	62.5	89	75.3	38	84.2
+2	19	73.7	85	94.1	44	81.8
+3	16	100.0	86	88.4	39	92.3
+4	47	80.9	132	84.8	39	87.2
Total	90	81.1	392	85.5	160	86.2

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